

TED(15) 3211
(Revision2015)

Reg.No-----
Signature-----

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY-
ANALOG CIRCUITS

Maximum marks 100

Time 3hours

PART A
(Maximum Marks:10)

- I Answer the following questions in one or two sentences. Each question carries 2 marks
- 1 List the methods of inter-stage coupling.
 - 2 What is meant by bandwidth of an amplifier?
 - 3 State the conditions for sustained oscillations.
 - 4 Draw the circuit diagram of an RC integrator circuit.
 - 5 What is slew rate of an OPAMP? (5x2=10)

PART B
(Maximum Marks:30)

- II Answer any five of the following questions. Each question carries 6 marks
- 1 Explain the importance of DC load-line of an amplifier.
 - 2 What are the advantages and disadvantages of an RC coupled amplifier?.
 - 3 List the features and applications of IC555.
 - 4 With the help of a circuit diagram explain the Current to voltage converter using Op Amp.
 - 5 List the applications of Clamping circuits.
 - 6 List the important electrical parameters of Op Amp.
 - 7 With the help of a circuit diagram and typical wave forms explain the working of Op-Amp voltage follower. (5x6=30)

PART -C
(Answer one full question from each module. Each question carries 15 marks.)

MODULE I

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| III | i | Calculate the voltage gain and input impedance of an CE amplifier. | 8 |
| | ii | Explain Direct coupling with a neat figure. | 7 |

OR

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| IV | i | What are the different schemes of feedback explain each with figure | 15 |
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MODULE II

V	i	Describe the working of an RC phase shift oscillator with a neat diagram.	8
	ii	Explain the working of a crystal oscillator	7

OR

VI	i	Derive the expression for frequency of oscillation of an astablemultivibrator using transistor	15
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MODULE III

VII	i	Explain the working of a positive series clipper with circuit diagram and wave form.	8
	ii	Explain the working of a RC high pass filter with circuit diagram.	7

OR

VIII	i	Explain the working of a RC low pass filter with circuit diagram.	8
	ii	Explain the working of a negative shunt clipper with circuit diagram and wave form.	7

MODULE IV

IX	I	Explain the working of a precision rectifier	8
	ii	Explain the input offset voltage and input bias- current of an Op-Amp	7

OR

		Explain the CMRR and input impedance of an Op-Amp	7
		Explain error detector circuit using Op-Amp	8